SIMATS ENGINEERING

CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**Sankuru Blessika**
Register Number	**192525251**

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Code of Conduct

I S. Blessika (192525251) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Blessika

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter Hosting Mission-Critical Applications. Analyze Performance, Security, and Manageability, then Justify the Better Choice.

Answer:

A hypervisor, also known as a Virtual Machine Monitor (VMM), is software that creates and manages multiple Virtual Machines (VMs) on a single physical server. Hypervisors enable virtualization by allowing several operating systems to share the same hardware resources efficiently. Hypervisors are classified into Type-1 (Bare Metal) and Type-2 (Hosted).

Comparison between Type-1 and Type-2 Hypervisors

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Runs directly on physical hardware	Runs on top of a host operating system

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Performance	High performance with low latency	Lower performance due to OS overhead
Security	More secure	Less secure
Resource Utilization	Efficient hardware utilization	Additional resources used by host OS
Reliability	Highly reliable	Depends on host OS stability
Scalability	Suitable for large datacenters	Suitable for testing and development
Examples	VMware ESXi, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation

Performance Analysis

Performance is critical in mission-critical applications such as banking systems, healthcare management, airline reservation systems, and e-commerce platforms.

- Type-1 hypervisors communicate directly with the hardware, eliminating unnecessary software layers.
- They provide better CPU scheduling, faster memory access, and lower disk I/O latency.
- They support advanced features such as live migration, high availability, and load balancing.
- Type-2 hypervisors must first communicate with the host operating system before accessing hardware, increasing latency and reducing overall performance.

Result: Type-1 Hypervisors provide significantly better performance than Type-2 Hypervisors.

Security Analysis

Security is extremely important because mission-critical applications store confidential and sensitive information.

Type-1 Hypervisor

- Smaller attack surface.
- No host operating system to exploit.
- Better isolation between virtual machines.
- Supports secure boot, encryption, and hardware-assisted virtualization.

Type-2 Hypervisor

- Depends on the security of the host operating system.
- Malware attacking the host OS can compromise all virtual machines.
- Larger attack surface.

Result: Type-1 Hypervisors provide stronger security.

Manageability Analysis

Modern cloud datacenters require centralized management.

Type-1 Hypervisors support:

- Live migration
- Automatic failover
- VM snapshots
- Resource scheduling
- High availability clusters
- Centralized monitoring
- Dynamic resource allocation

Type-2 Hypervisors mainly provide basic virtualization and are suitable for desktop environments.

Justification

For mission-critical cloud applications, Type-1 Hypervisor is the best choice because it offers:

- Highest performance

- Better security
- Higher reliability
- Efficient resource utilization
- Better scalability
- Enterprise-level management
- Minimal downtime
- High availability

2. A Cloud Provider Must Isolate Workloads Belonging to Finance, Healthcare, and Student Portals on the Same Hardware. Analyze How Virtualization Architecture Supports Isolation and Identify Two Major Attack Surfaces.

Answer:

Cloud service providers often host multiple customers on the same physical server. Finance, healthcare, and educational applications contain sensitive data, making workload isolation essential. Virtualization architecture provides logical separation between these workloads while sharing the same hardware.

How Virtualization Supports Isolation

A hypervisor creates multiple independent virtual machines (VMs), where each VM has:

- Its own operating system
- Dedicated virtual CPU
- Allocated memory
- Virtual disk storage
- Virtual network interface

Although the VMs share the same physical server, they cannot directly access each other's resources.

Isolation Mechanisms

1. CPU Isolation

Each VM receives allocated processor cores and scheduling time, preventing one workload from affecting another.

2. Memory Isolation

Each VM has its own memory space. Memory protection prevents unauthorized access between VMs.

3. Storage Isolation

Separate virtual disks ensure that finance, healthcare, and student data remain isolated.

4. Network Isolation

Virtual LANs (VLANs), virtual switches, and software-defined networking separate network traffic between workloads.

Benefits of Virtualization Isolation

- Improved security
- Better resource utilization
- Fault isolation
- Compliance with regulations
- Easier maintenance
- Flexible scaling

Major Attack Surface 1: Hypervisor Attack

The hypervisor controls all virtual machines. If attackers exploit a vulnerability in the hypervisor, they may gain control over every VM running on the server.

Protection

- Regular patching
- Secure boot
- Strong administrator authentication
- Hardware-assisted virtualization

Major Attack Surface 2: VM Escape

VM escape occurs when malicious software running inside one VM breaks out of its isolated environment and accesses the hypervisor or another VM.

Protection

- Updated hypervisors
- Access control policies
- Network segmentation
- Intrusion detection systems

Additional Security Measures

- Encryption of stored data
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Continuous monitoring
- Security audits

3. Study the Following VM Host Utilization Snapshot and Recommend Corrective Actions: CPU 92%, Memory 88%, Storage I/O Latency 35 ms, Average VM Boot Time 170 s. Explain the Likely Bottlenecks.

Answer:

The given utilization statistics indicate that the VM host is experiencing resource exhaustion and poor performance.

Host Statistics

Resource	Current Value
CPU Utilization	92%
Memory Utilization	88%
Storage I/O Latency	35 ms
Average VM Boot Time	170 seconds

Analysis of CPU Usage

CPU utilization of **92%** is extremely high.

Effects

- Slow execution of applications
- Delayed VM scheduling
- Increased response time
- Reduced throughput

Bottleneck

The processor is overloaded.

Analysis of Memory Usage

Memory utilization of **88%** leaves very little free RAM.

Effects

- Memory swapping

- Increased paging
- Slow application performance
- Reduced VM responsiveness

Bottleneck

Insufficient physical memory.

Analysis of Storage I/O Latency

Storage latency of **35 ms** is much higher than recommended values.

Effects

- Slow disk read/write operations
- Delayed application startup
- Slow database performance
- Long VM boot time

Bottleneck

Storage subsystem is overloaded.

Analysis of VM Boot Time

Average VM boot time is **170 seconds**, which is considerably high.

Possible reasons:

- High CPU load
- Storage bottleneck
- Memory pressure

Corrective Actions

CPU Optimization

- Add more CPU cores.
- Migrate some VMs to another host.
- Enable load balancing.

Memory Optimization

- Increase RAM.
- Optimize memory allocation.
- Remove idle VMs.

Storage Optimization

- Replace HDDs with SSD/NVMe drives.
- Use storage caching.
- Optimize storage configuration.

General Improvements

- Monitor resource utilization continuously.
- Use automatic scaling.
- Schedule workload balancing.
- Upgrade hardware if necessary.

4. Analyze How Software Defined Datacenter (SDDC) Architecture Improves Agility Compared with a Traditional Datacenter. Support Your Answer with Compute, Storage, and Network Virtualization Considerations.

Answer:

A **Software Defined Datacenter (SDDC)** is a modern data center architecture where compute, storage, networking, and security are fully virtualized and managed through software. Unlike a traditional datacenter that depends heavily on physical hardware and manual configuration, an SDDC automates resource provisioning and management.

Traditional Datacenter vs. Software Defined Datacenter

Traditional Datacenter	Software Defined Datacenter (SDDC)
Hardware-based infrastructure	Software-controlled infrastructure
Manual provisioning	Automated provisioning
Fixed resource allocation	Dynamic resource allocation
Slow deployment	Rapid deployment
Limited scalability	Elastic scalability
Higher operational cost	Lower operational cost

1. Compute Virtualization

Compute virtualization allows multiple virtual machines (VMs) to run on a single physical server.

Advantages

- Better hardware utilization
- Faster VM provisioning
- Easy workload migration
- High availability
- Efficient resource sharing

2. Storage Virtualization

Storage virtualization combines multiple physical storage devices into a single logical storage pool.

Advantages

- Simplified storage management
- Dynamic storage allocation
- Snapshots and backups
- Disaster recovery support
- Better storage utilization

3. Network Virtualization

Network virtualization creates logical networks independent of physical hardware using technologies such as Software-Defined Networking (SDN).

Advantages

- Faster network configuration
- Secure traffic isolation
- Centralized network management
- Easy scalability
- Improved network performance

How SDDC Improves Agility

- Rapid deployment of new services
- Automated provisioning and management
- Dynamic scaling of resources
- Reduced manual intervention
- Faster disaster recovery
- Improved resource utilization
- Lower operational costs

- Better flexibility for cloud environments

5. A Multi-Cloud Federation Project Fails Due to Identity Mismatches Between Providers. Analyze the Issue and Propose a Secure Identity and Privacy Design Using Federation Concepts.

Answer:

A multi-cloud environment allows organizations to use services from multiple cloud providers. However, different providers often use different identity management systems, leading to identity mismatches that prevent users from accessing resources seamlessly.

Problem Analysis

Identity mismatches occur because:

- Different authentication protocols are used.
- User identities are stored in separate directories.
- Access control policies differ between providers.
- User attributes may not be compatible across systems.

As a result, users authenticated in one cloud may not be recognized by another cloud, causing authentication failures and poor user experience.

Secure Identity Design

1. Identity Federation

A trusted **Identity Provider (IdP)** authenticates users once and shares trusted identity information with all cloud providers.

2. Single Sign-On (SSO)

Users log in once and can securely access multiple cloud platforms without repeated authentication.

3. Federation Standards

Use widely accepted standards such as:

- **SAML (Security Assertion Markup Language)** for enterprise authentication.
- **OAuth 2.0** for secure authorization.
- **OpenID Connect (OIDC)** for modern identity verification.

4. Multi-Factor Authentication (MFA)

Require users to verify their identity using:

- Password
- OTP
- Biometric authentication
  This reduces the risk of unauthorized access.

5. Role-Based Access Control (RBAC)

Grant permissions based on user roles (e.g., administrator, doctor, student, finance staff), ensuring users access only the resources they need.

6. Privacy Protection

- Encrypt identity data during transmission and storage.
- Share only the minimum required user attributes.
- Maintain detailed audit logs for accountability.
- Regularly review access permissions.

Benefits of the Proposed Design

- Seamless authentication across multiple cloud providers.
- Improved security and user privacy.
- Reduced administrative overhead.
- Consistent access control policies.
- Better compliance with organizational and regulatory requirements.